

1. Objective:

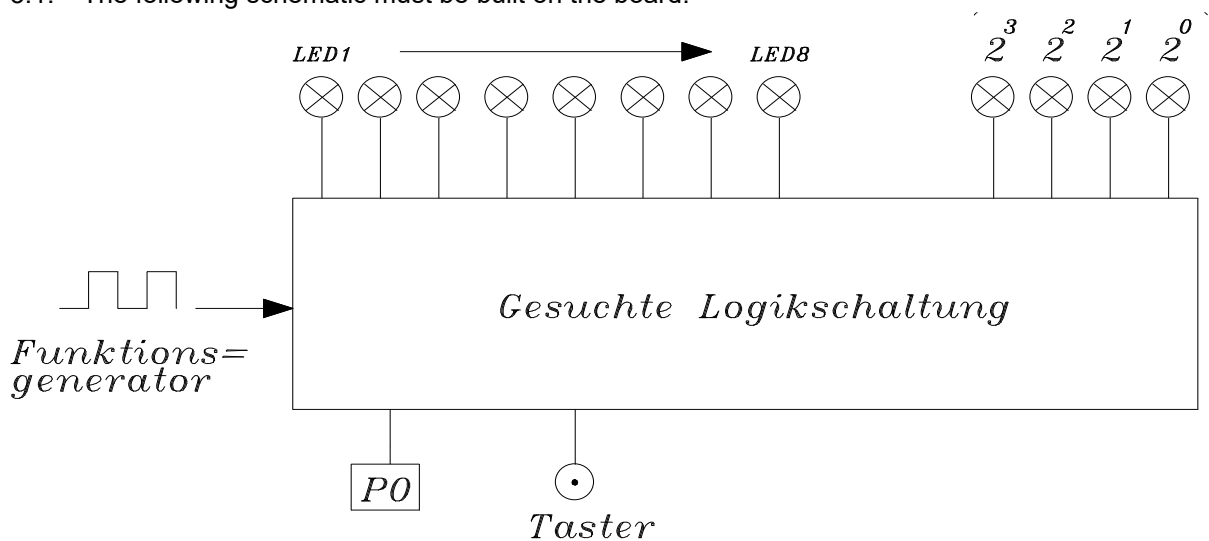
Understand the mode of operation of sequential circuits like shift register and counter.

2. Preparation:

- 2.1. Prepare the schematics for all tasks using the file "Lab_02_SRC_Guide.pdf" which can be found in the same directory in ILIAS and present them to the tutor before starting the lab exercise.
- 2.2. The schematic can and should be simulated using LTSpice, following the hints given in the file "Lab_02_SRC_Simulation" which can be found in the same directory in ILIAS.

3. Tasks:

- 3.1. The following schematic must be built on the board:



PO = Power-On-Reset

In a row of 8 LEDs only a single LED shall be ON. First LED1 shall be ON. On the next rising edge of the signal input from the waveform generator LED2 shall get ON and LED1 OFF. This shall continue until LED8 gets OFF and then restart with LED1. After power-on and reset the LED1 should be the first to be ON.

- 3.2. Augment the first schematic to "catch" the light. Whenever LED5 is ON and a push-button is pressed, the LED5 shall stay ON for 2 seconds, and then the light shall continue to "move".
- 3.3. Augment the second schematic by adding a decimal counter and 4 LEDs. Every time the light is "caught", the counter shall count, and this shall be displayed in BCD code with the 4 LEDs.

For all these tasks, a square-wave signal with a frequency of approx. 1 Hz, an amplitude of 5 V_{pp} and an offset of 2.5 V must be used. The circuit must be supplied with 5 V DC.

4. Report:

Not requested.

5. Notes:

The following integrated circuits can be used:

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|---|---|
| 1x 74HC00 (4x 2In-NAND-Gate) | 1x 74HC160 (1x Synchronous 4-Bit Counter) |
| 1x 74HC74 (2x D-Type Flip-Flop) | 1x 74HC164 (1x 8-Bit Shift Register) |
| 1x 74HC132 (4x 2In-NAND-Gate Schmitt-Trigger) | 1x 74HC221 (2x Monoflop) |